Binary Numbers — Short Revision Notes

Evaluate Level: Make Judgments and Justify Decisions

1. Verifying Conversions — Decimal to Binary

Problem	Student's Answer	Verification	Judgment
13_{10} → Binary	1101_2	13÷2=6R1, 6÷2=3R0, 3÷2=1R1, 1÷2=0R1 → 1101_2	CORRECT
22_{10} → Binary	10110_2	22÷2=11R0, 11÷2=5R1, 5÷2=2R1, 2÷2=1R0, 1÷2=0R1 → 10110_2	CORRECT
15_{10} → Binary	1111_2	15÷2=7R1, 7÷2=3R1, 3÷2=1R1, 1÷2=0R1 → 1111_2	CORRECT
8_{10} → Binary	1000_2	8÷2=4R0, 4÷2=2R0, 2÷2=1R0, 1÷2=0R1 → 1000_2	CORRECT
5_{10} → Binary	101_2	5÷2=2R1, 2÷2=1R0, 1÷2=0R1 → 101_2	CORRECT
6_{10} → Binary	110_2	6÷2=3R0, 3÷2=1R1, 1÷2=0R1 → 110_2	CORRECT
10_{10} → Binary	1010_2	10÷2=5R0, 5÷2=2R1, 2÷2=1R0, 1÷2=0R1 → 1010_2	CORRECT
17_{10} → Binary	10001_2	17÷2=8R1, 8÷2=4R0, 4÷2=2R0, 2÷2=1R0, 1÷2=0R1 → 10001_2	CORRECT

Common Mistakes to Judge:

Mistake	Why Wrong	Correct
$13_{10} = 1110_2$	13÷2 results: 1,1,0,1	1101_2 ✓
$6_{10} = 011_2$	Leading zeros not needed	110_2 ✓
$22_{10} = 10100_2$	Missing last remainder	10110_2 ✓

2. Verifying Conversions — Binary to Decimal

Problem	Student's Answer	Verification	Judgment
1101_2 → Decimal	13_{10}	1×8+1×4+0×2+1×1 = 13	CORRECT
10110_2 → Decimal	22_{10}	1×16+0×8+1×4+1×2+0×1 = 22	CORRECT
1111_2 → Decimal	15_{10}	1×8+1×4+1×2+1×1 = 15	CORRECT
1000_2 → Decimal	8_{10}	1×8+0×4+0×2+0×1 = 8	CORRECT
1010_2 → Decimal	12_{10}	1×8+0×4+1×2+0×1 = 10, not 12	INCORRECT
1100_2 → Decimal	12_{10}	1×8+1×4+0×2+0×1 = 12	CORRECT
1110_2 → Decimal	14_{10}	1×8+1×4+1×2+0×1 = 14	CORRECT
10001_2 → Decimal	17_{10}	1×16+0×8+0×4+0×2+1×1 = 17	CORRECT

Common Mistakes to Judge:

Mistake	Why Wrong	Correct
$1010_2 = 12_{10}$	8+0+2+0 = 10, not 12	10_{10} ✓
$1110_2 = 13_{10}$	8+4+2+0 = 14, not 13	14_{10} ✓
$1001_2 = 10_{10}$	8+0+0+1 = 9, not 10	9_{10} ✓

Verification Method:

- Write place values (8, 4, 2, 1)
- Multiply each digit by its place value
- Add the results
- Compare with the given answer

3. Judging the Reasonableness of a Binary Number

Is the Binary Number Reasonable?

Binary	Decimal	Reasonable ?	Justification
101_2	5_{10}	YES	5 is between 4 and 7 (3-digit binary numbers)
111_2	7_{10}	YES	7 is the maximum with 3 digits
1000_2	8_{10}	YES	8 is the first 4-digit number
1111_2	15_{10}	YES	15 is the maximum with 4 digits
10001_2	17_{10}	YES	17 is between 16 and 31 (5-digit numbers)
11001_2	25_{10}	YES	25 is between 16 and 31 (5-digit numbers)
11111_2	31_{10}	YES	31 is the maximum with 5 digits

Reasonableness Checklist:

Check 1: Number of digits must match the range
- 1 digit: 0 to 1
- 2 digits: 2 to 3
- 3 digits: 4 to 7
- 4 digits: 8 to 15
- 5 digits: 16 to 31

Check 2: First digit must be 1 (except for 0)
- $0_2 = 0$
- $1_2 = 1$
- $10_2 = 2$

Check 3: All digits must be 0 or 1
- 102_2 is INVALID (contains 2)

4. Evaluating Binary Addition

Problem	Student's Answer	Verification	Judgment
$101_2 + 10_2$	111_2	101_2=5, 10_2=2, 5+2=7=111_2	CORRECT
$110_2 + 1_2$	111_2	110_2=6, 1=1, 6+1=7=111_2	CORRECT
$1011_2 + 111_2$	10010_2	11+7=18, 10010_2=18	CORRECT
$111_2 + 111_2$	1110_2	7+7=14, 1110_2=14	CORRECT
$1010_2 + 101_2$	1111_2	10+5=15, 1111_2=15	CORRECT
$1111_2 + 1_2$	10000_2	15+1=16, 10000_2=16	CORRECT
$101_2 + 11_2$	1000_2	5+3=8, 1000_2=8	CORRECT
$1101_2 + 1010_2$	10111_2	13+10=23, 10111_2=23	CORRECT

Common Mistakes to Judge:

Mistake	Why Wrong	Correct
$111_2 + 1_2 = 110_2$	7+1=8 (1000_2) not 6	1000_2 ✓
$101_2 + 101_2 = 1000_2$	5+5=10 (1010_2) not 8	1010_2 ✓
$111_2 + 111_2 = 1110_2$	7+7=14 (1110_2)	CORRECT ✓

Verification Method:

- Convert both binary numbers to decimal
- Add the decimal numbers
- Convert the sum back to binary
- Compare with the student's answer

5. Evaluating Binary Subtraction

Problem	Student's Answer	Verification	Judgment
$110_2 - 1_2$	101_2	$6–1=5=101_2$	**CORRECT**
$100_2 - 1_2$	11_2	$4–1=3=11_2$	**CORRECT**
$1010_2 - 10_2$	1000_2	$10–2=8=1000_2$	**CORRECT**
$1100_2 - 11_2$	1001_2	$12–3=9=1001_2$	**CORRECT**
$1000_2 - 11_2$	101_2	$8–3=5=101_2$	**CORRECT**
$1111_2 - 101_2$	1010_2	$15–5=10=1010_2$	**CORRECT**
$10101_2 - 11_2$	10010_2	$21–3=18=10010_2$	**CORRECT**

Common Mistakes to Judge:

Mistake	Why Wrong	Correct
$100_2 - 1_2 = 1_2$	$4–1=3=11_2$, not 1	11_2 ✓
$110_2 - 1_2 = 100_2$	$6–1=5=101_2$, not 4	101_2 ✓
$1010_2 - 10_2 = 110_2$	$10–2=8=1000_2$, not 6	1000_2 ✓

Verification Method:

- Convert both binary numbers to decimal
- Subtract the decimal numbers
- Convert the difference back to binary
- Compare with the student's answer

6. Evaluating the Correct Base

Is the Number Written Correctly?

Number	Correct?	Why?
101_2	YES	Valid binary (digits 0, 1 only)
102_2	NO	Digit 2 not allowed in base 2
110_2	YES	Valid binary
112_2	NO	Digit 2 not allowed in base 2
1111_2	YES	Valid binary
1010_2	YES	Valid binary
100_2	YES	Valid binary
101 (with subscript 10)	NO	Wrong base for binary, should be subscript 2

Evaluation Checklist:

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Check 1: Are all digits 0 or 1?
- If yes → Valid binary
- If no → Invalid binary

Check 2: Is the subscript correct?
- Binary: subscript 2
- Decimal: subscript 10
- 101_2 (correct for binary)
- 101_10 (correct for decimal)

Check 3: Are leading zeros needed?
- 00101_2 is the same as 101_2
- Leading zeros are not needed
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7. Evaluating the Number of Digits

Binary Number	Number of Digits	Correct?	Justification
1_2	1	YES	1 digit needed for 1
10_2	2	YES	2 digits needed for 2
100_2	3	YES	3 digits needed for 4
1000_2	4	YES	4 digits needed for 8
1111_2	4	YES	4 digits needed for 15
10000_2	5	YES	5 digits needed for 16

Rule:

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Number of binary digits = position of highest power of 2 + 1

For 13_10 (1101_2):
Highest power = 2^3 = 8
Digits = 3 + 1 = 4 ✓
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For 22₁₀ (10110₂):
Highest power = 2⁴ = 16
Digits = 4 + 1 = 5 ✓
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8. Evaluating Complex Additions

Problem	Student's Answer	Verification	Judgment
$11101_2 + 1101_2$	101010_2	$29+13=42=101010_2$	CORRECT
$1110_2 + 111_2$	10101_2	$14+7=21=10101_2$	CORRECT
$11011_2 + 11_2$	11110_2	$27+3=30=11110_2$	CORRECT
$100111_2 + 11_2 + 1_2$	101011_2	$39+3+1=43=101011_2$	CORRECT
$11110_2 + 1110_2 + 110_2$	101010_2	$30+14+6=50=110010_2$, not 42	INCORRECT

Common Mistakes:

Mistake	Why Wrong	Correct
$11110_2+1110_2+110_2 = 101010_2$	$30+14+6=50=110010_2$, not 42	110010_2 ✓
$1011_2+111_2 = 10010_2$	$11+7=18=10010_2$ (Correct)	✓

9. Evaluating Complex Subtractions

Problem	Student's Answer	Verification	Judgment
$11001_2 - 111_2$	10010_2	$25-7=18=10010_2$	CORRECT
$10001_2 - 101_2$	1100_2	$17-5=12=1100_2$	CORRECT
$11111_2 - 111_2$	11000_2	$31-7=24=11000_2$	CORRECT
$110011_2 - 1100_2 - 110_2$	10001_2	$51-12-6=33=100001_2$, not 17	INCORRECT

Common Mistakes:

Mistake	Why Wrong	Correct
$110011_2-1100_2-110_2 = 10001_2$	$51-12-6=33=100001_2$	100001_2 ✓

10. Evaluation Checklist

When verifying a conversion:

- Is the division method used correctly?
- Are remainders written in the correct order?
- Are place values used correctly for binary to decimal?
- Is the result reasonable?

When verifying addition:

- Are columns added from right to left?
- Are carries handled correctly?
- Does $1+1=10_2$ (not 2)?
- Does $1+1+1=11_2$ (carry 1)?
- Convert to decimal to verify?

When verifying subtraction:

- Is borrowing handled correctly?
- When 0–1, borrow 1 from left
- Borrowed 1 = 2 in current column
- Convert to decimal to verify?

When evaluating a binary number:

- Are all digits 0 or 1?
- Is the subscript correct (2)?
- Is the number of digits reasonable?
- Is the leading digit 1 (except for 0)?

When checking reasonableness:

- Does the decimal value match the binary?
- Is the value in the correct range?
- Does the arithmetic make sense?

11. Self-Test — Evaluation Questions

No.	Question	Answer	Justification
1	Is $13_{10} = 1101_2$ correct?	YES	13÷2=6R1, 6÷2=3R0, 3÷2=1R1, 1÷2=0R1 → 1101_2
2	Is $22_{10} = 10100_2$ correct?	NO	$22=10110_2$ (not 10100_2)
3	Is $1101_2 = 13_{10}$ correct?	YES	8+4+0+1=13
4	Is $111_2 + 1_2 = 1000_2$ correct?	YES	$7+1=8=1000_2$
5	Is $100_2 - 1_2 = 11_2$ correct?	YES	$4–1=3=11_2$
6	Is 102_2 a valid binary number?	NO	Binary uses only 0 and 1

No.	Question	Answer	Justification
7	Is $101_2 + 10_2 = 111_2$ correct?	**YES**	$5+2=7=111_2$
8	Is $1110_2 = 14_{10}$ correct?	**YES**	$8+4+2+0=14$
9	Is $10101_2 - 11_2 = 10010_2$ correct?	**YES**	$21-3=18=10010_2$
10	Is 110_{10} a binary number?	**NO**	Subscript 10 means decimal, not binary

12. Quick Evaluation Guide

■ CORRECT if:

- ■ All digits are 0 or 1
- ■ Subscript is 2 for binary (e.g., 101_2)
- ■ Decimal → Binary uses repeated division by 2
- ■ Binary → Decimal uses powers of 2
- ■ $1+1=10_2$ (carry 1)
- ■ $1+1+1=11_2$ (carry 1)
- ■ Subtraction: borrow 1 = 2 in current column
- ■ Leading zeros are not required

■ INCORRECT if:

- ■ Contains digits other than 0 or 1
- ■ Wrong subscript (e.g., 101_{10} for binary)
- ■ 1+1=0 without carry (forgetting carry)
- ■ 0–1 without borrowing
- ■ $13_{10} = 1110_2$ (should be 1101_2)
- ■ $1010_2 = 12_{10}$ (should be 10_{10})

13. Common Errors to Judge

Error	Why Wrong	Correct
$13_{10} = 1110_2$	Wrong division result	$13_{10} = 1101_2$
$1010_2 = 12_{10}$	8+0+2+0=10, not 12	10_{10}
$111_2 + 1_2 = 110_2$	7+1=8 (1000_2)	1000_2
$100_2 - 1_2 = 1_2$	4–1=3 (11_2)	11_2
102_2 (digit 2)	Binary only has 0 and 1	Invalid
101_{10} as binary	Wrong subscript	Should be 101_2

Revision Tip: *Always verify your answers by converting back to decimal. This is the easiest way to check if your binary calculations are correct! If the decimal values match, your binary work is correct.*